

Text Document

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Test Cases

Project Title

****Flood Prediction System Using Machine Learning****

Test Case Table

Test ID	Test Case	Input	Expected Result
	Actual Result	Status	
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TC-01	Launch the Flask application	Run `python app.py`	Application starts successfully without errors
			Application started successfully
			Pass
TC-02	Open the dashboard	Open `http://127.0.0.1:5000` in a browser	Flood Prediction Dashboard is displayed
			Dashboard displayed correctly
			Pass
TC-03	Load Machine Learning model	Load `flood_model.pkl`	Model loads successfully
			Model loaded successfully
			Pass
TC-04	Enter valid input values	Valid disaster details	Prediction is generated
			Prediction generated successfully
			Pass
TC-05	Submit the prediction form	Click **Predict Flood**	Prediction result is displayed
			Prediction displayed correctly
			Pass
TC-06	Display flood risk percentage	Valid input values	Risk percentage is calculated and shown
			Risk percentage displayed
			Pass
TC-07	Display Doughnut Chart	Prediction completed	Doughnut chart appears
			Chart displayed successfully
			Pass
TC-08	Display Bar Chart	Prediction completed	Bar chart appears
			Chart displayed successfully
			Pass
TC-09	Leave a required field empty	Missing rainfall value	Browser

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prevents form submission	Validation message displayed	Pass	
TC-10 Enter decimal values	Latitude, Longitude, Rainfall	System	
accepts decimal values	Accepted successfully	Pass	
TC-11 Enter large numeric values	Large rainfall or affected population	System	
processes data correctly	Prediction generated	Pass	
TC-12 Refresh the webpage	Browser refresh		
Dashboard reloads without errors	Reloaded successfully	Pass	
TC-13 Check responsive layout	Open on different screen sizes	Interface	
adjusts properly	Responsive layout verified	Pass	
TC-14 Verify animations	Open dashboard		
Background, waves, and rain animations work	Animations displayed correctly	Pass	
TC-15 Verify dataset loading	Load `flood_dataset_classification.csv`	Dataset	
loads successfully	Dataset loaded successfully	Pass	

Summary

* **Total Test Cases:** 15
* **Passed:** 15
* **Failed:** 0
* **Overall Result:** **PASS**

Conclusion

All planned test cases were executed successfully. The Flood Prediction System correctly loaded the dataset and trained model, accepted valid user inputs, generated flood predictions, displayed the calculated flood risk percentage, and visualized the results using charts. Input validation prevented incomplete form submissions, and the user interface remained responsive with all animations functioning correctly. Based on the executed test cases, the application meets the expected functional and performance requirements.

